Machine Learning: The Engine of Modern AI

Rule-Based vs. Machine Learning

Traditional programming is explicit: a human writes every rule. Machine Learning (ML) flips the model — instead of writing rules, you provide labelled examples and let the algorithm discover its own statistical rules.

Traditional Programming	Machine Learning
Human writes rules	Algorithm learns rules from data
Input + Rules = Output	Input + Output = Rules (learned)
Brittle — breaks on new cases	Generalises to unseen cases
Example: spam keyword filter	Example: spam classifier trained on emails

> *Core insight: ML inverts the programming paradigm — data + desired outputs train the rules, not the other way around.*

The Three Types of Machine Learning

Supervised Learning: Labelled data (inputs + correct answers). Algorithm learns to map inputs to outputs. Examples: spam detection, image classification, price prediction.

Unsupervised Learning: Unlabelled data. Algorithm finds hidden patterns or clusters. Examples: customer segmentation, anomaly detection, recommendation systems.

Reinforcement Learning: Agent takes actions in an environment to maximise a cumulative reward. Examples: game-playing (AlphaGo), robotics, self-driving car steering.

The Training Pipeline

A typical supervised ML workflow:

- 1. Collect and label data (the most expensive step).
- 2. Split into training set and test/validation set.
- 3. Choose a model architecture (decision tree, neural net, etc.).
- 4. Train: feed data, calculate error (loss), adjust parameters.
- 5. Evaluate on unseen test data. Iterate until performance is satisfactory.
- 6. Deploy the model to production.

Overfitting vs. Underfitting

Overfitting: Model memorises the training data but fails on new data (too complex).

Underfitting: Model is too simple to capture patterns even in training data.

Goldilocks zone: Good generalisation — performs well on both training and test data.

> *Analogy: A student who memorises past exam papers but can't answer new questions is 'overfitting'. One who hasn't studied enough is 'underfitting'.*

Key Takeaways

- ML learns statistical rules from labelled data instead of being explicitly programmed.
- Three paradigms: Supervised, Unsupervised, and Reinforcement Learning.
- The training pipeline: collect data → train → evaluate → deploy.
- Overfitting and underfitting are the two key failure modes to watch for.

Quick-Revision Questions

- Contrast rule-based programming with ML using a concrete example.
- Which type of ML would you use to group customers by purchase behaviour? Why?
- What is overfitting and how can you detect it?